Session 2 — Statistics & Inference

Solution sheet

Complete worked solutions, with reminders of the key ideas. Numerical answers are boxed. Variances here use the population form (divide by n); the sample form (divide by $n-1$) is noted where relevant.

Exercise 1 *Firing rates* $6, 8, 10, 12, 14$.

Solution. Mean $= \dfrac{6+8+10+12+14}{5} = \dfrac{50}{5} = \boxed{10}$. Deviations $-4, -2, 0, 2, 4$; squared $16, 4, 0, 4, 16$ summing to 40. So variance $= \dfrac{40}{5} = \boxed{8}$ and SD $= \sqrt{8} = \boxed{\approx 2.83}$.

> **Reminder**
> Variance is the *average squared deviation* from the mean; the SD returns to the original units. (Sample variance would divide by $n-1=4$, giving 10.)

Exercise 2 *Scores mean* 70*, SD* 8.

Solution.

(a) $z = \dfrac{86-70}{8} = \boxed{2}$.

(b) Standardised scores always have mean $\boxed{0}$ and SD $\boxed{1}$.

(c) $Y = 1.5X + 10$: mean $= 1.5(70) + 10 = \boxed{115}$; SD $= |1.5| \times 8 = \boxed{12}$.

> **Reminder**
> Under $Y = aX + b$: the mean transforms as $a\bar{X} + b$, but the SD scales by $|a|$ only — *adding a constant never changes the spread.*

Exercise 3 $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5)$; $\bar{x} = 3, \bar{y} = 4$.

Solution. Products $(x-\bar{x})(y-\bar{y})$: $4, 0, 0, 0, 2$, summing to 6.

(a) $\mathrm{Cov}(x, y) = \dfrac{6}{5} = \boxed{1.2}$.

(b) $\sum(x-\bar{x})^2 = 10$, $\sum(y-\bar{y})^2 = 6$, so $r = \dfrac{6}{\sqrt{10 \cdot 6}} = \dfrac{6}{\sqrt{60}} = \boxed{\approx 0.775}$.

(c) A fairly strong *positive* association: the two neurons tend to be high or low together.

Reminder

Correlation is covariance divided by the two SDs, which makes it unit-free and bounded in $[-1, 1]$. (The r value does not depend on whether you divide the covariance by n or $n-1$ — the factor cancels.)

Exercise 4 *Single-trial noise* $\sigma = 12$.

Solution.

(a) $\mathrm{SE} = \dfrac{12}{\sqrt{36}} = \dfrac{12}{6} = \boxed{2}$.

(b) $\dfrac{12}{\sqrt{n}} = 1 \Rightarrow n = \boxed{144}$.

(c) Since $\mathrm{SE} \propto 1/\sqrt{n}$, precision improves only with the *square root* of the effort — diminishing returns.

Reminder

Standard error of the mean $= \sigma/\sqrt{n}$. To cut it by a factor k you need k^2 times as much data.

Exercise 5 $n = 64$, $\bar{x} = 540$, $\sigma = 80$.

Solution. $\mathrm{SE} = \dfrac{80}{\sqrt{64}} = \dfrac{80}{8} = 10$.

(a) 95%: $540 \pm 1.96(10) = 540 \pm 19.6 = \boxed{[520.4,\ 559.6]}$.

(b) 99%: $540 \pm 2.576(10) = 540 \pm 25.76 = \boxed{[514.2,\ 565.8]}$.

(c) The 99% interval is wider: demanding more confidence requires casting a wider net.

Reminder

A confidence interval is $\bar{x} \pm z\ \mathrm{SE}$. Higher confidence $\Rightarrow$ larger multiplier z (1.96 for 95%, 2.576 for 99%) $\Rightarrow$ wider interval.

Exercise 6 44 *of* 60 *correct; under* H_0 *mean* 30, *SD* $\sqrt{15} \approx 3.87$.

Solution. $z = \dfrac{44 - 30}{3.87} = \dfrac{14}{3.87} = \boxed{\approx 3.61}$. Since $|z| \gg 1.96$, the result is highly significant: the subject performs well above chance.

Reminder

A z-test compares the observed count to what H_0 predicts, in units of the null standard deviation. $|z| > 1.96$ corresponds to $p < 0.05$ (two-sided).

Exercise 7 *Patients* 520, *controls* 500, *SD* 40, $n = 25$ *each.*

Solution.

(a) $d = \dfrac{520 - 500}{40} = \boxed{0.5}$ (a "medium" effect).

(b) $\mathrm{SE}_{\mathrm{diff}} = 40\sqrt{\frac{1}{25} + \frac{1}{25}} = 40\sqrt{0.08} \approx 11.31$; $t = \dfrac{20}{11.31} = \boxed{\approx 1.77}$.

(c) $|t| < 1.96$, so *not* significant at 5% — despite a genuine medium effect. The study is simply underpowered ($n = 25$ per group).

Reminder

Effect size (d) and statistical significance (t, p) answer different questions. A real effect can be missed when the sample is small: significance needs both a real effect *and* enough data.

Exercise 8

Solution.

(a) **Type I**: concluding patients and controls differ when they do not (a false positive). **Type II**: failing to detect a real difference (a false negative).

(b) Increasing n raises the test's *power*, which most directly reduces $\boxed{\text{Type II}}$ errors.

Reminder

Power $= 1 - \beta$ is the chance of detecting a real effect. Bigger samples, larger effects, and less noise all increase power.

Exercise 9

$x = (0, 1, 2, 3, 4)$, $y = (1, 1, 2, 2, 4)$; $\bar{x} = \bar{y} = 2$.

Solution. Products $(x - \bar{x})(y - \bar{y})$: $2, 1, 0, 0, 4$ summing to 7; $\sum(x - \bar{x})^2 = 10$.

(a) $b = \dfrac{7}{10} = \boxed{0.7}$; $\quad a = \bar{y} - b\bar{x} = 2 - 0.7(2) = \boxed{0.6}$. Line: $\hat{y} = 0.6 + 0.7x$.

(b) $\hat{y}(5) = 0.6 + 0.7(5) = \boxed{4.1}$.

(c) $R^2 = r^2 = 0.90^2 = \boxed{\approx 0.82}$: the line explains about 82% of the variance in firing rate.

Reminder

The least-squares slope is $b = \mathrm{Cov}(x, y)/\,\mathrm{Var}(x)$, the line always passes through $(\bar{x}, \bar{y})$, and $R^2 = r^2$ is the fraction of variance "explained".

Exercise 10

Averaging $n = 100$ die rolls (single roll: mean 3.5, SD ≈ 1.71).

Solution.

(a) $\mathbb{E}[\bar{X}] = \boxed{3.5}$ (the sample mean is unbiased).

(b) $\text{SE} = \dfrac{1.71}{\sqrt{100}} = \boxed{0.171}$.

(c) Approximately $\boxed{\text{Normal}}$, by the Central Limit Theorem.

Reminder

CLT: for large n, $\bar{X} \approx \mathcal{N}(\mu,\ \sigma^2/n)$ *whatever* the shape of the original distribution — here, a flat die distribution becomes a bell curve after averaging.

Exercise 11 *Two noise sources, variances 9 and 16.*

Solution.

(a) Independent $\Rightarrow$ variances add: $\text{Var} = 9 + 16 = \boxed{25}$, $\text{SD} = \boxed{5}$.

(b) Perfectly correlated $\Rightarrow$ SDs add: $3 + 4 = \boxed{7}$ (variance 49).

Reminder

$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X, Y)$. Independence makes $\text{Cov} = 0$; perfect correlation makes $\text{Cov} = \text{SD}_X\,\text{SD}_Y$, so the SDs add.

Exercise 12 *130 on test A ($\mu = 100, \sigma = 15$) vs. 60 on test B ($\mu = 50, \sigma = 5$).*

Solution. $z_A = \dfrac{130 - 100}{15} = 2.0$ and $z_B = \dfrac{60 - 50}{5} = 2.0$. The two z-scores are $\boxed{\text{equal}}$: the participant is 2 SD above the mean on *both*, so their relative standing is the same.

Reminder

z-scores put different scales on a common footing — the only fair way to compare a score of 130 with a score of 60.

Exercise 13 *Simpson's paradox.*

Solution.

(a) Within each severity group, $\boxed{\text{A}}$ is better: $93\% > 87\%$ (mild) and $73\% > 69\%$ (severe).

(b) Pooled, $\boxed{\text{B}}$ looks better: $83\% > 78\%$.

(c) This is **Simpson's paradox**. Treatment A was given mostly to the harder (severe) cases and B mostly to the easier (mild) ones, so the different case mix reverses the aggregate comparison.

Reminder

A lurking variable (here, case severity) can reverse a trend when data are pooled. Always ask whether groups are comparable before combining them.

Exercise 14 *Regression to the mean.*

Solution.

(a) An extreme low score reflects both true (low) ability *and* bad luck on the day. On a retest the luck component averages out, so the score drifts back up toward the mean — even with no teaching.

(b) Because the programme enrolled only the lowest scorers, some improvement is guaranteed by this effect alone. Without a control group you cannot tell how much (if any) is due to the programme.

Reminder

Groups selected for being extreme will, on remeasurement, move toward the mean. Randomised control groups protect you from mistaking this for a real effect.

Exercise 15 95% *CI for a difference* $= [2, 8]$ *ms.*

Solution.

(a) 0 lies *outside* $[2, 8]$, so the difference $\boxed{\text{is significant}}$ at the 5% level.

(b) The true difference is a fixed (unknown) number, not random — so it does not have a "95% probability" of being anywhere. Correct reading: the *method* produces intervals that capture the true value in 95% of repeated studies.

Reminder

"The 95% CI excludes 0" is exactly equivalent to "significant at the 5% level". The confidence statement is about the long-run procedure, not this one interval.